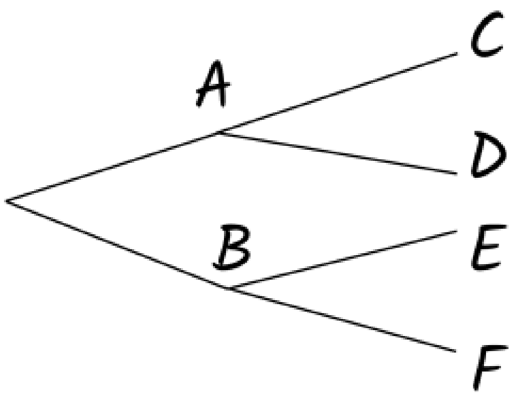
Name: __________________________ Exit quiz

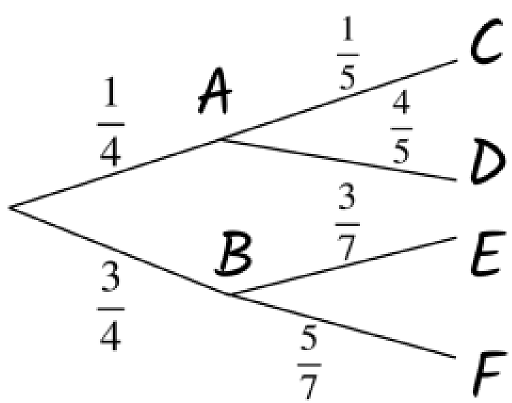
Calculating theoretical probabilities from probability trees (two events)

1 If P(A) = $\frac{1}{3}$ and P(A and D) = $\frac{1}{5}$, then what is P(D)? Tick **1** correct answer



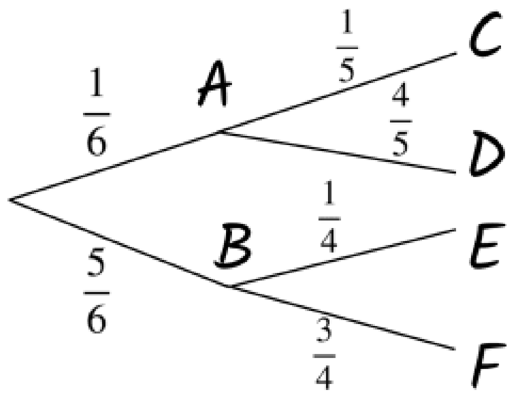
- [ ] $\frac{3}{5}$
- [ ] $\frac{1}{5}$
- [ ] $\frac{2}{5}$

2 What is the error in this tree diagram? Tick **1** correct answer



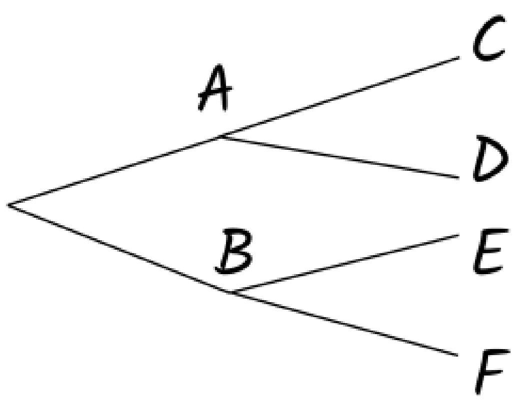
- [ ] Events B and F should sum to 1 but they don't.
- [ ] Events A and C should sum to 1 but they don't.
- [ ] Events D and E should sum to 1 but they don't.
- [ ] Events E and F should sum to 1 but they don't.

3 What is a suitable calculation for the probability of event B occurring, and event F not occurring? Tick **1** correct answer



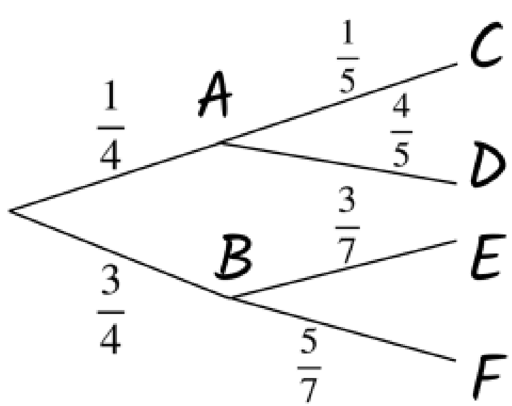
- [ ] $\frac{5}{6} * \frac{1}{4}$
- [ ] $\frac{5}{6} + \frac{1}{4}$
- [ ] $\frac{5}{6} * \frac{3}{4}$
- [ ] $\frac{5}{6} + \frac{3}{4}$

4 If the first and second event in this tree diagram were both flipping a coin once, what would be the probability of event B and event F occurring? Tick **1** correct answer



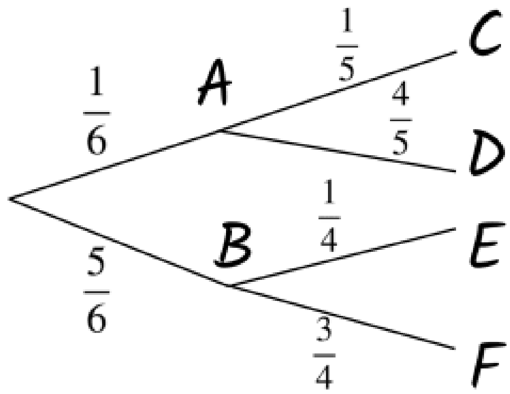
- [ ] $\frac{1}{4}$
- [ ] $\frac{1}{2}$
- [ ] $\frac{2}{6}$

5 What is the least likely (but not impossible) pair of events occurring in this tree diagram? Tick **1** correct answer



- [ ] A then C
- [ ] A then D
- [ ] B then E
- [ ] B then F

6 Which pair of events has a more than even chance of occurring? Tick **1** correct answer



- [ ] B and F
- [ ] B and E
- [ ] A and C
- [ ] A d D